Global CO_2 Emissions

Who Emits, How Much, and What's Changing

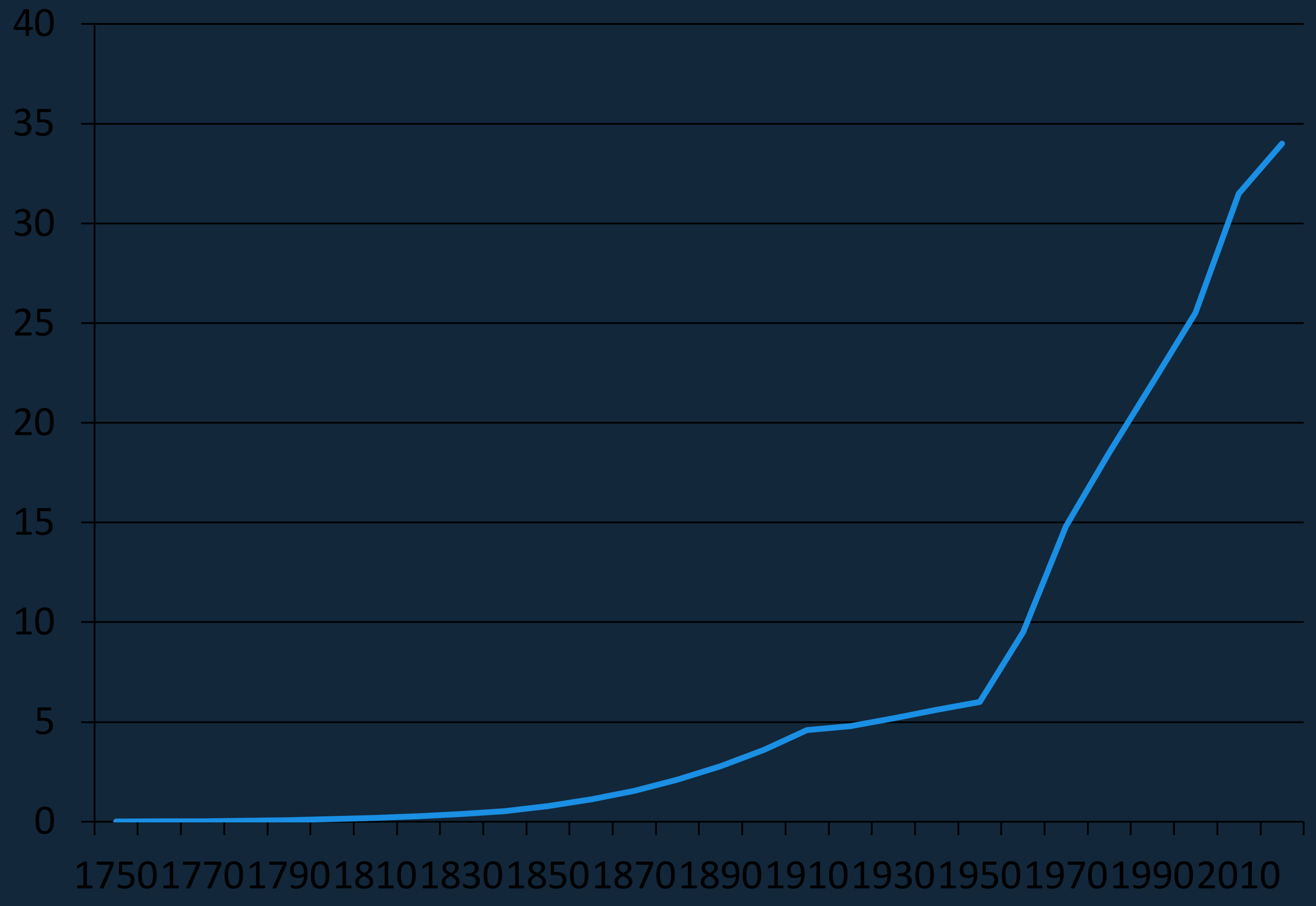
Scope: Fossil fuels & industry · 1750–2024
Data: Global Carbon Budget 2025 via Our World in Data
Authors: Hannah Ritchie & Max Roser

For policy analysts · climate negotiators · sustainability researchers

>35 Bt	~31%	150×	1750
annual CO_2 (2024)	China's share	per capita gap	data start year

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Fossil fuels & industry · 1750–2024



Pre-1900

Emissions negligible; coal drives first growth

1950 → 6 Bt

Post-WWII industrial expansion accelerates

1990 → 22 Bt

Nearly 4× growth since 1950

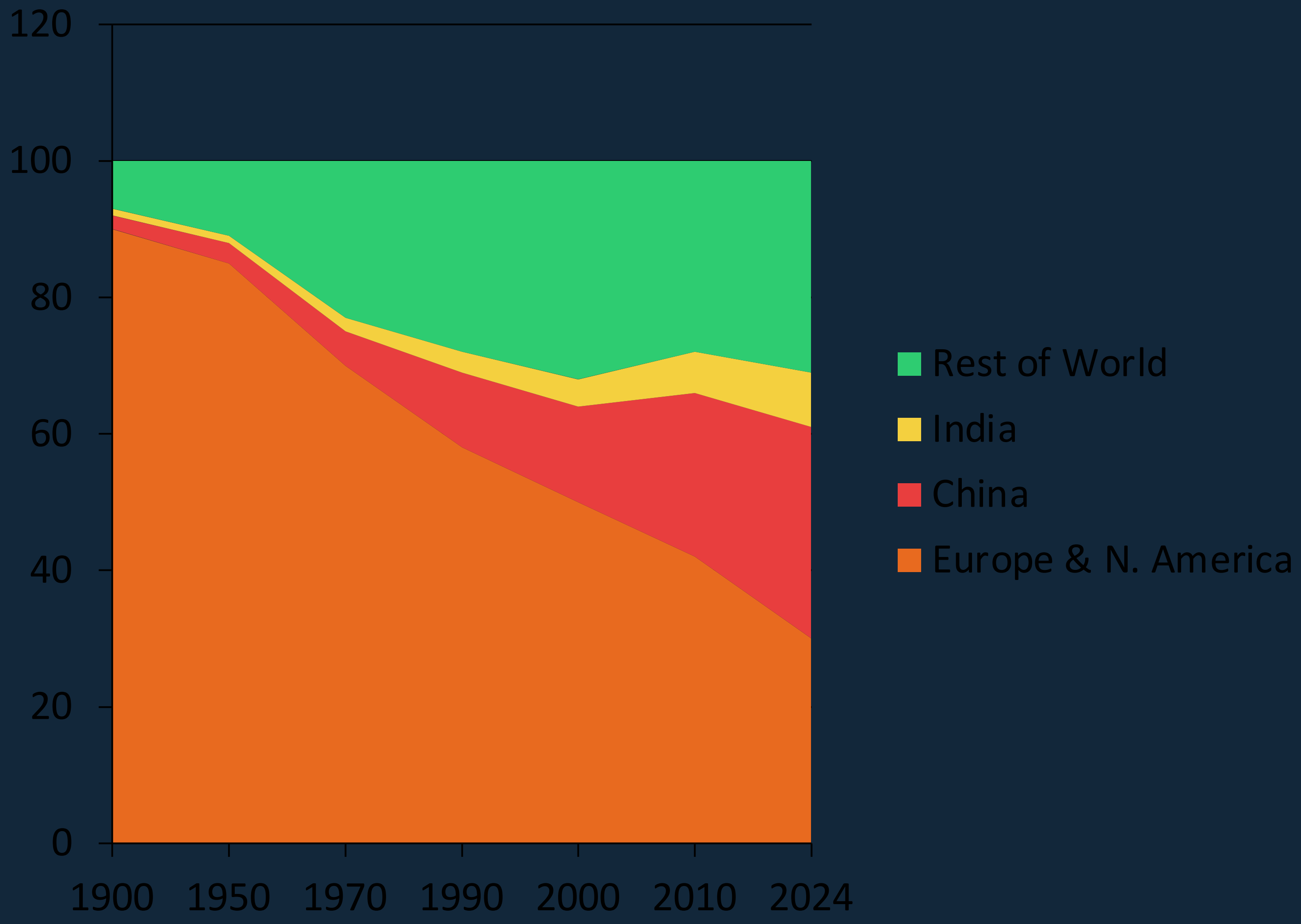
2024 → 35+ Bt

Growth slowing but no peak reached yet

Source: Global Carbon Budget 2025 · Our World in Data

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

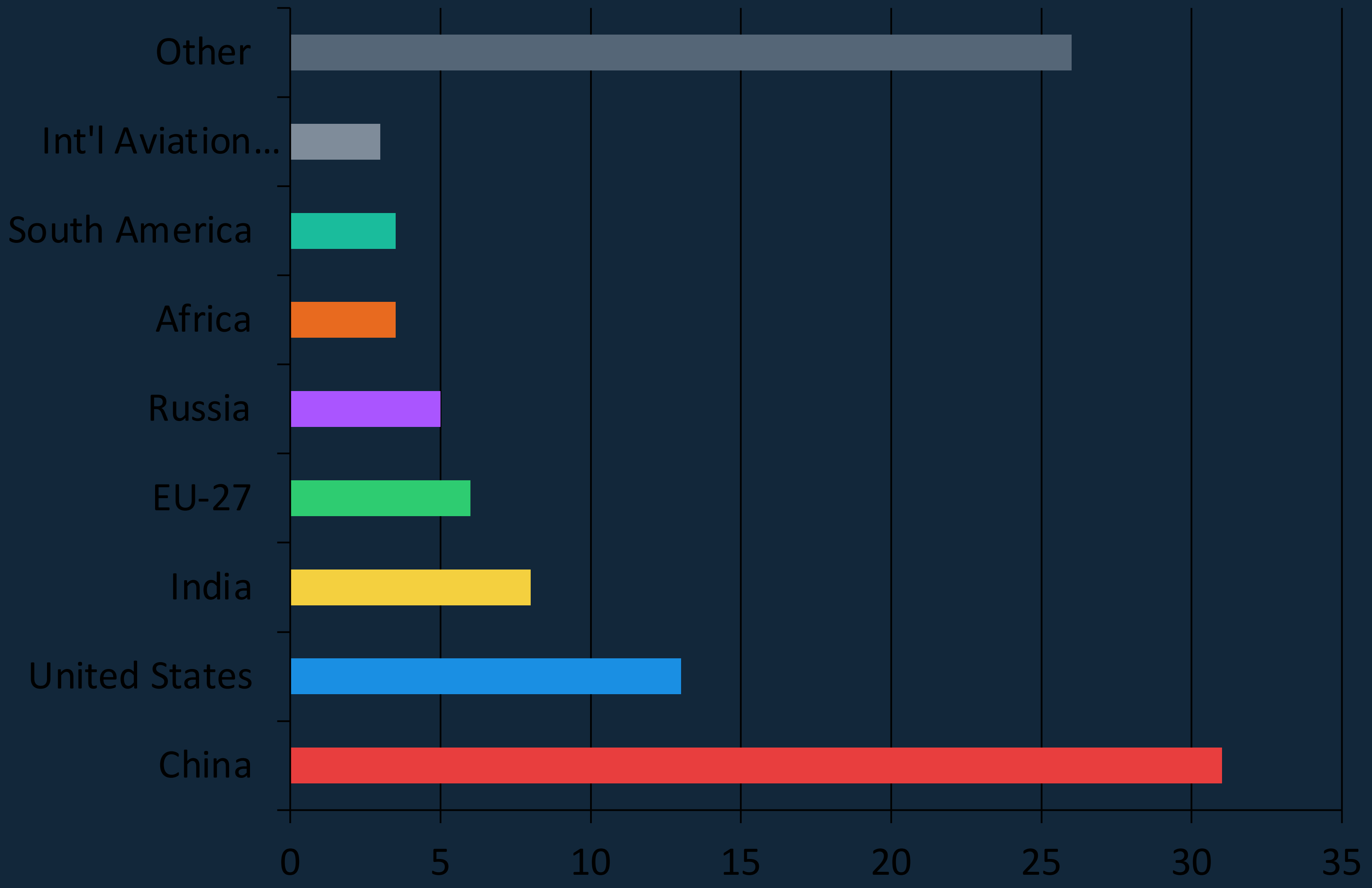
Share of global CO_2 emissions · 1900–2024



1900	Europe + US	>90%
1950	Europe + US	>85%
2024	Europe + US	<33%
2024	China alone	~31%
2024	Asia total	~50%

China Now Emits More Than One-Quarter of Global CO_2

Share of global CO_2 emissions · 2024



Key insight

China (~31%) now emits more than the EU-27 and US combined.

US trajectory

The US (~13%) has been declining steadily.

India rising

India (~8%) is the fastest-growing major emitter.

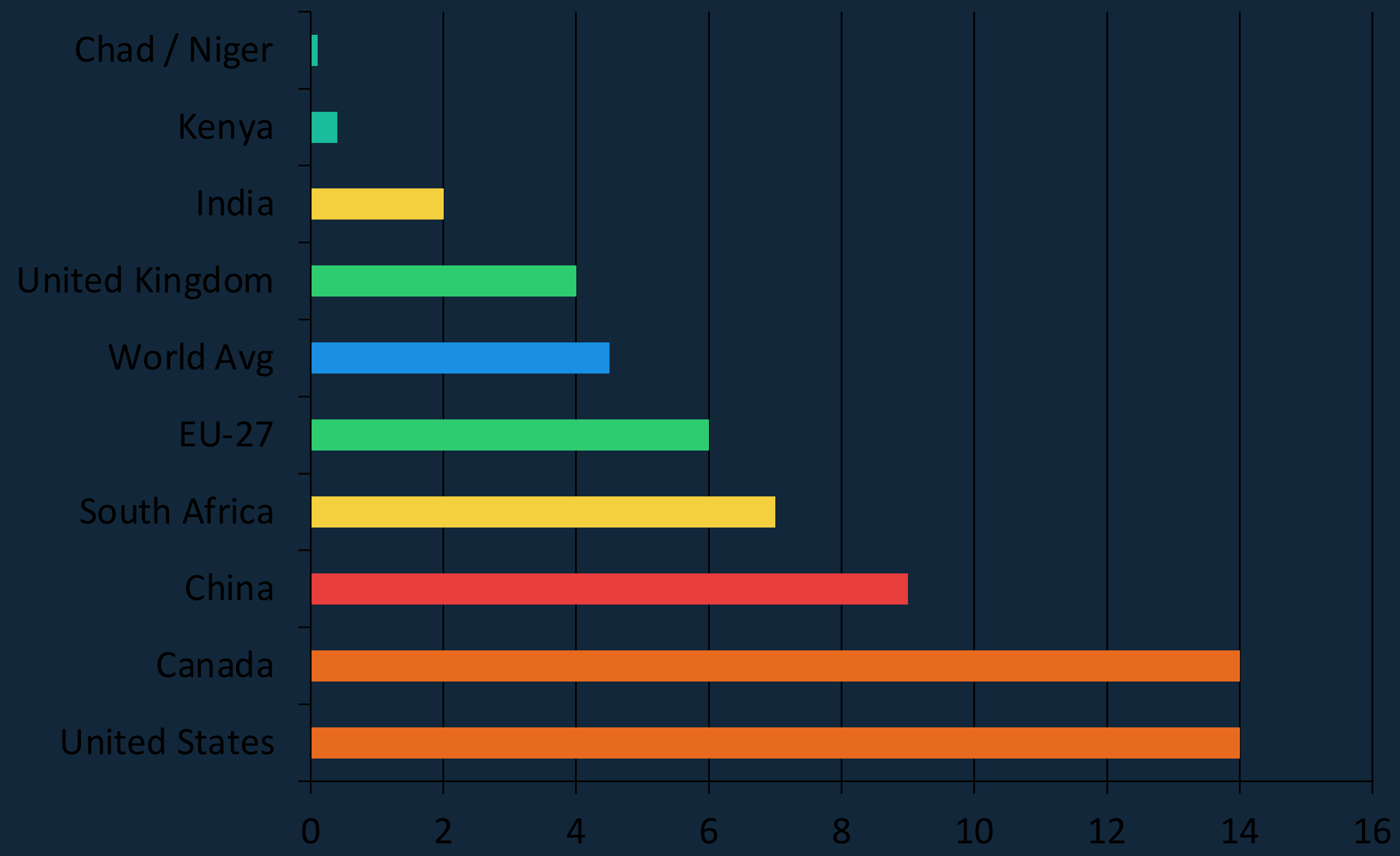
Africa / S. Am.

Each ~3–4% — comparable to int'l aviation & shipping.

Source: Global Carbon Budget 2025 · Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

Per capita CO_2 · selected countries · approx. 2024



150×

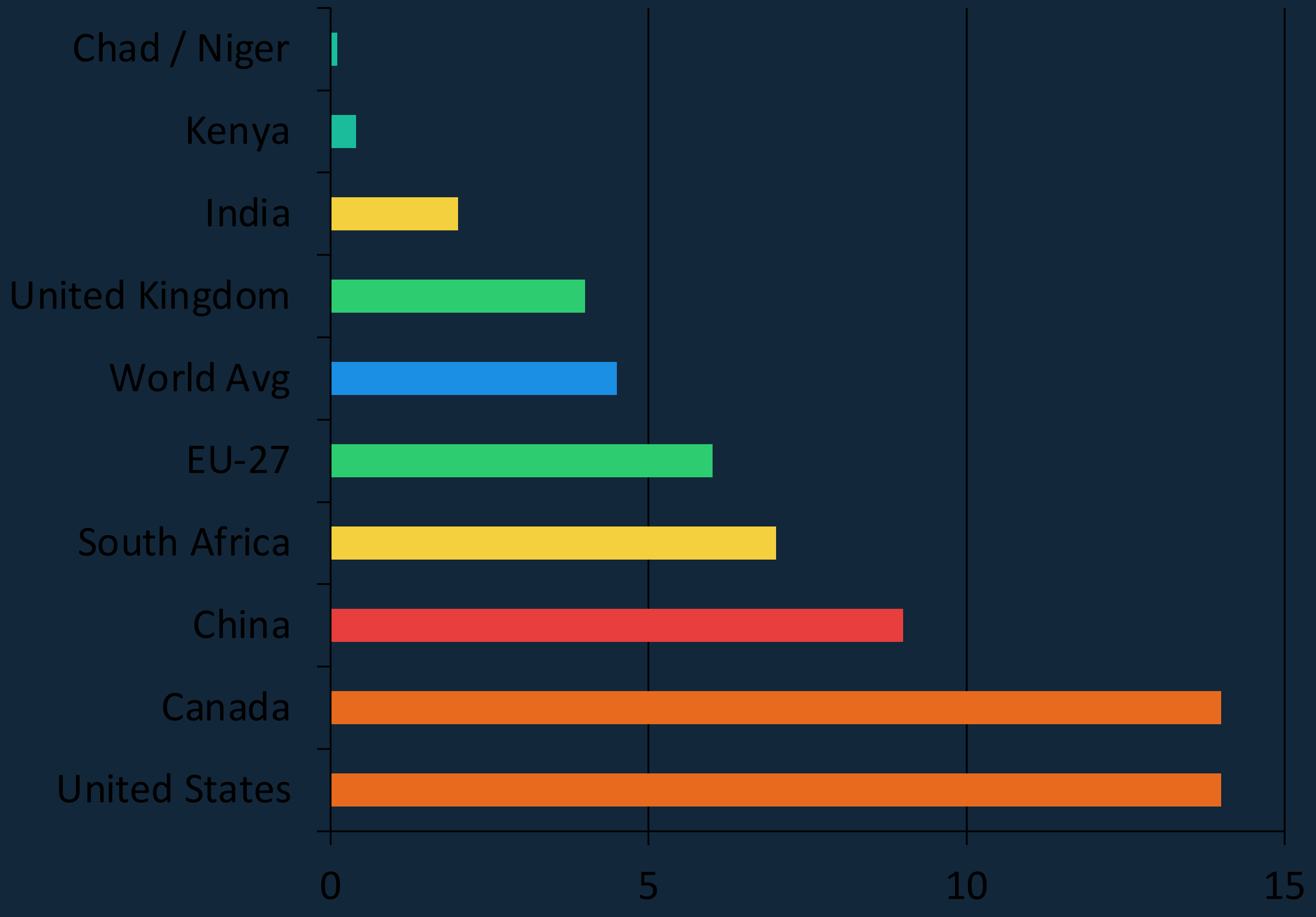
gap between Chad/Niger and the US/Canada

Among wealthy nations: France & UK (~4 t) emit far less than Germany or US (~6–14 t) due to nuclear and renewable energy mix.

Source: Global Carbon Budget 2025 · Our World in Data

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Per capita CO_2 · key countries · 2024 snapshot



US & Canada ~14 t:

3× the global average of ~4.5 t

China ~9 t:

Below US despite being the largest total emitter

India ~2 t:

Very low per capita; large population drives total share

EU-27 ~6 t:

Declining, boosted by renewables and efficiency

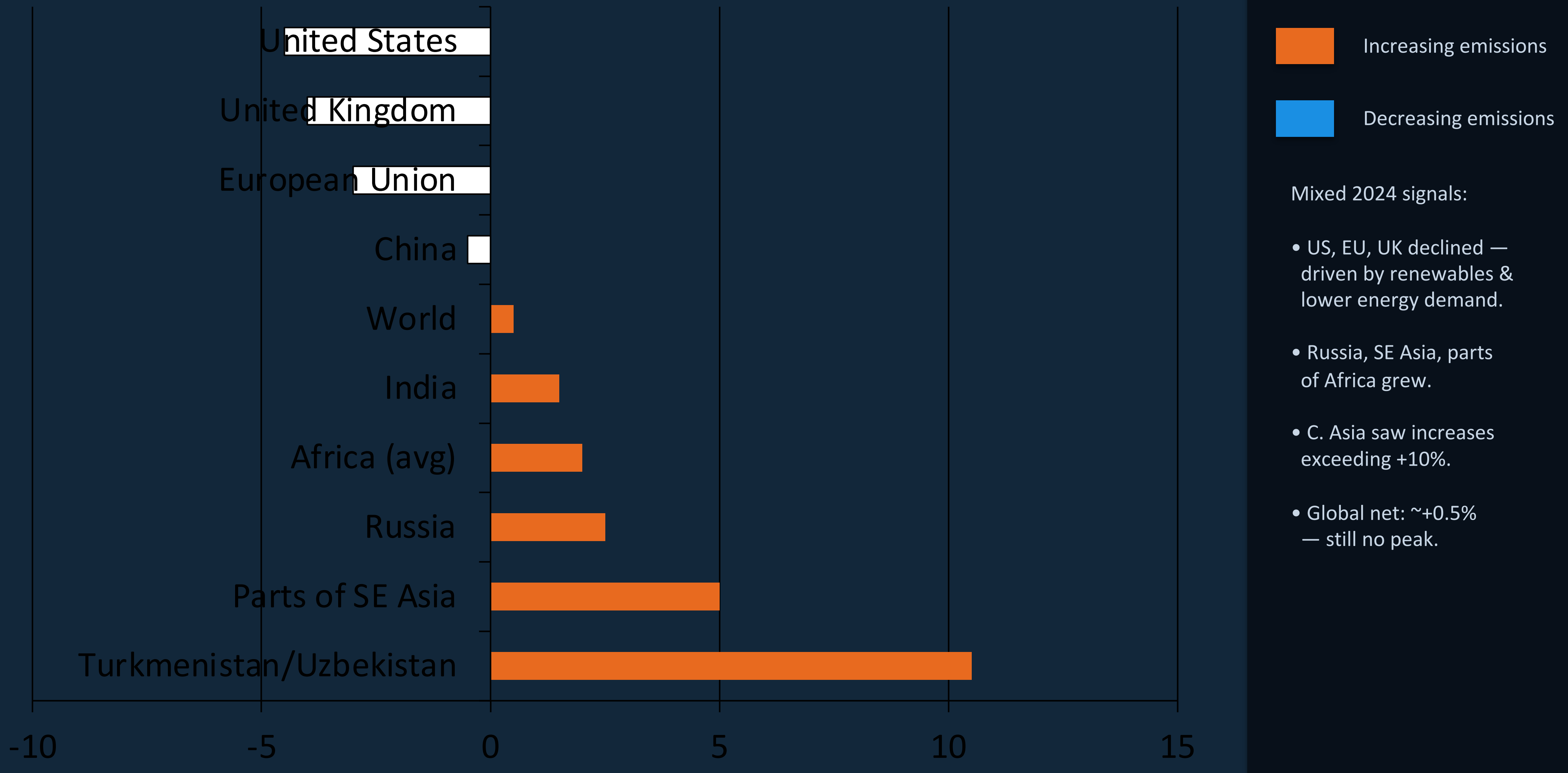
Kenya ~0.4 t:

Sub-Saharan average; ~35× below the US

Source: Global Carbon Budget 2025 · Our World in Data · figure_8.jpg for historical trends

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Annual % change in CO_2 emissions · 2024



Mixed 2024 signals:

- US, EU, UK declined — driven by renewables & lower energy demand.
- Russia, SE Asia, parts of Africa grew.
- C. Asia saw increases exceeding +10%.
- Global net: ~+0.5% — still no peak.

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

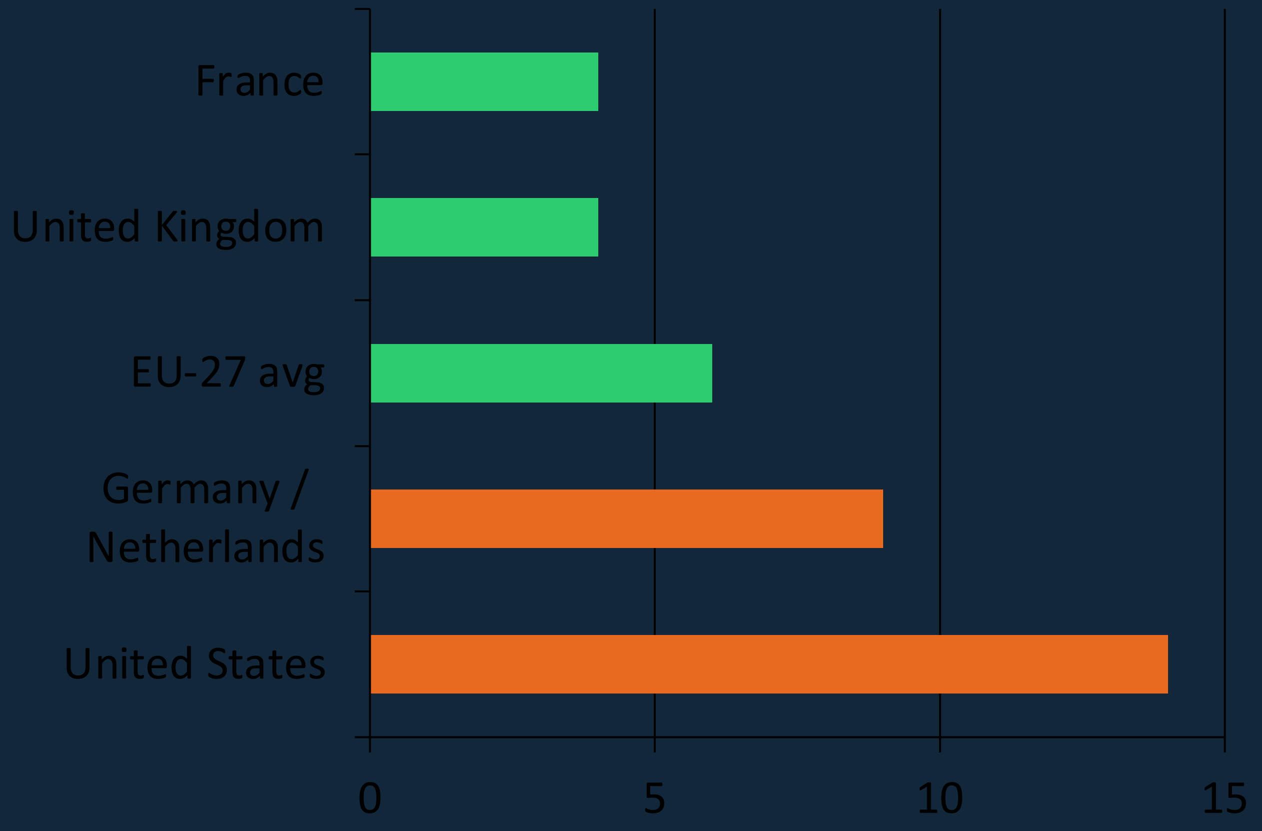
Per capita CO_2 · wealthy nations · energy context

Same prosperity, different emissions

Among high-income nations, per capita CO_2 emissions vary dramatically — not because of wealth differences, but because of energy mix choices:

- France and the UK have extensive nuclear and renewable capacity, lowering their electricity-sector emissions.
- Germany and the Netherlands have historically relied more on coal and natural gas, pushing per capita figures higher.
- The United States combines high energy consumption with a fossil-fuel-heavy grid — despite recent renewables growth.

This shows that prosperity alone does not determine emissions: policy, technology, and infrastructure choices matter enormously.



Orange = fossil-fuel-heavy grid   Green = nuclear/renewables-heavy grid

Source: Global Carbon Budget 2025 · Our World in Data

Key Takeaways: Responsibility Is Shared but Unequal

Summary · Global CO_2 emissions 1750–2024

1 **No peak yet**

Global CO_2 from fossil fuels exceeds 35 Bt/yr (2024) and has not yet peaked, despite slowing growth. Cumulative concentrations continue to rise.

2 **China is the largest annual emitter — but not per capita**

China (~31% of global total) emits ~9 t/person — well below the US (~14 t). Total vs. per capita framing leads to very different policy conclusions.

3 **Historical responsibility dominated by Europe and the US**

Europe and the US accounted for >90% of emissions in 1900 and >85% in 1950. Today they account for <33%. Cumulative historical emissions remain heavily skewed.

4 **A 150× per capita gap persists**

The average American or Canadian emits ~150× more than someone in Chad or Niger. Africa as a whole contributes only 3–4% of global emissions.

5 **Energy policy can decouple prosperity from emissions**

France and the UK emit ~4 t/person vs. ~14 t in the US — at similar living standards. Nuclear and renewable energy investment is a proven lever.